PSYC1071: Introduction to Psychology 1
Animal Learning and Cognition

Lecture 1:
Historical background

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Learn Ultra:

↳ Introduction to Psychology 1:

↳ Course Documents

↳ Block 3: Animal Learning and Cognition

Lecture slides, overviews, revision materials

Professor Anthony McGregor

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If you don't understand something:

- Ask

(or)

- Listen again to the lecture when it uploads to Lecture Recordings
- Read the lecture notes I post after the lecture
- If you still don't follow then post your question on the Discussion Board

Contact: Office hour 10-11am Tuesdays . Email for appointment outside this time

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My ‘lecture style’

- I tend not to write many words on my slides. I find this stops me reading them, but it also means my lectures are a bit more ‘free form’.
- My rambles into trivia are often give you a break from concentrating on some difficult content. It’s all part of the plan.
- Apart from some obvious fact-based material, this is not a lecture series to learn by rote. Sit back and listen. Don’t try to write down what I say word-for-word.

Why Study Animal Cognition?

- Comparative Psychology
  - How do different species differ in their psychological abilities/processes?
- Natural and intellectual curiosity
- Relevance to other disciplines
  - human cognitive psychology
  - behavioural neuroscience
  - artificial intelligence and computational modelling
  - clinical psychology

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Psychology is a science in which behavioural and other evidence is used to understand the internal processes leading people (and members of other species) to behave as they do (Eysenck, 2004)

Darwin (1859)
On the Origin of Species by Natural Selection

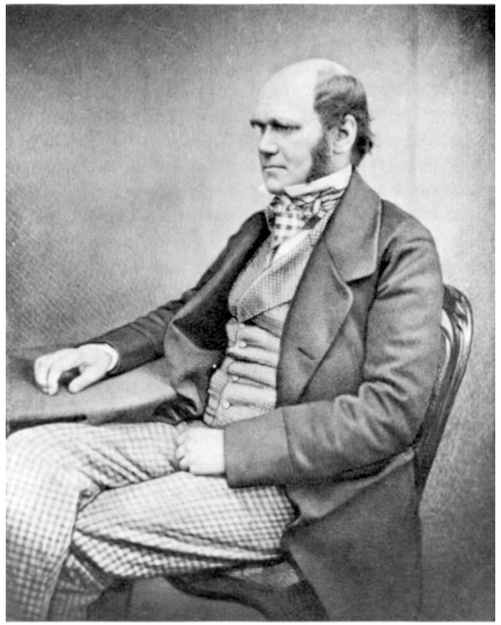


Fig. 1.2. Charles Darwin at about the time of writing *The Origin of Species*

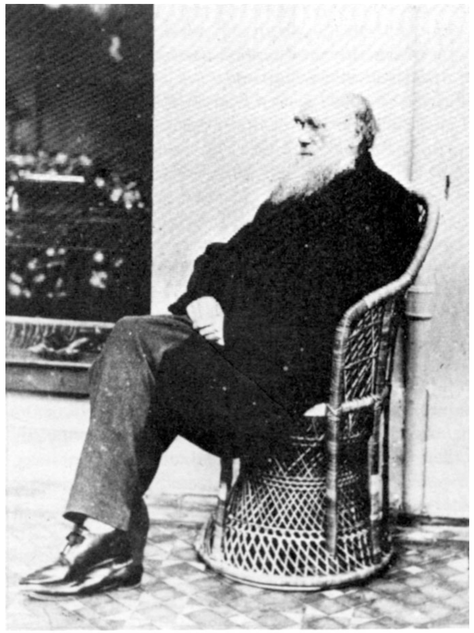
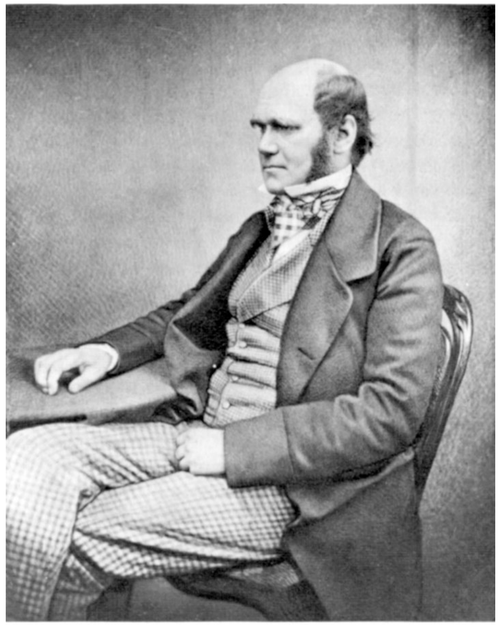


Fig. 2.1 Charles Darwin at about the time of meeting Romanes

Darwin (1859)
On the Origin of Species by Natural Selection



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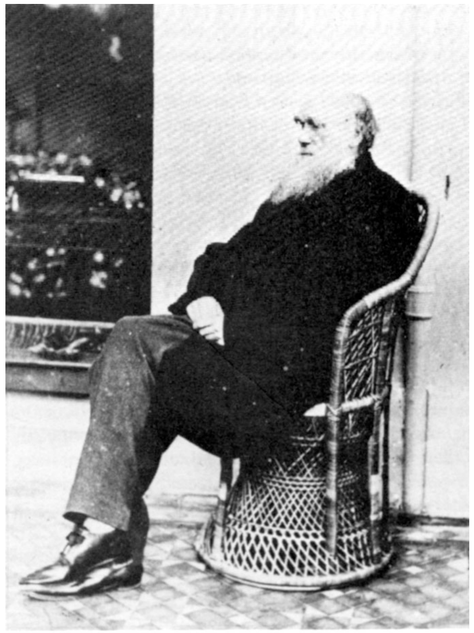

Fig. 2.1 Charles Darwin at about the time of meeting Romanes

- there is variability among individuals in a population
- variability is hereditary
- more offspring are born than can survive, and there is competition among individuals for resources
- the survivors of the competition will reproduce

Darwin (1859)
On the Origin of Species by Natural Selection

- There is no difference between the evolution of humans and the evolution of other animals
- There is no difference between behaviour and other evolved traits

- there is variability among individuals in a population
- variability is hereditary
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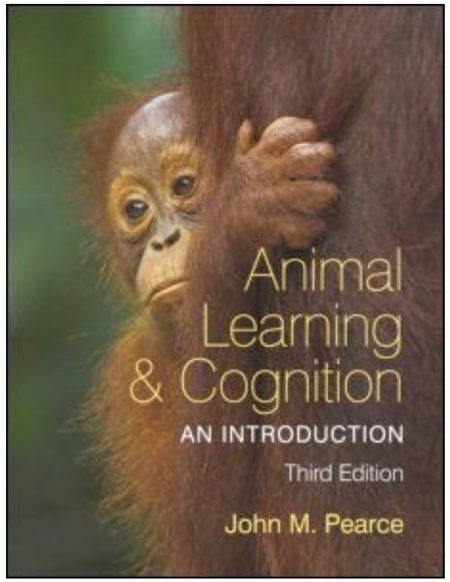
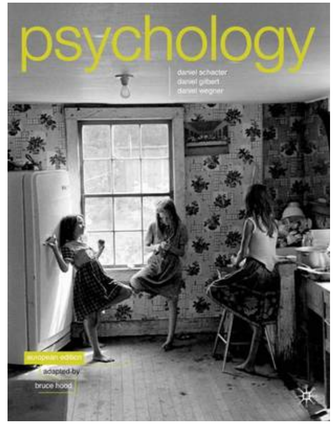
Suggested Textbook

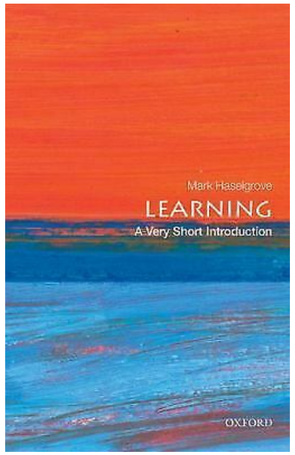


"There are very few instances in which one of the world's leading researchers is willing to write a textbook in his area of expertise, and even fewer where the resulting text is beautifully clear and interesting. Pearce's text is of just this quality; any student fortunate enough to read it will be exposed to a lucid, entertaining, and stimulating introduction to what we know about animal learning and cognition." - **Dr. David Lieberman, Department of Psychology, University of Stirling**

Suggested Textbook



General introduction to learning. Covers basics, useful background



Mark Haselgrove - Learning: A very short introduction

Very short, but covers a lot of what's in our lectures and more. Cheap!

https://www.amazon.co.uk/Learning-Very-Short-Introduction-Introductions/dp/0199688362

Historical Background

Romanes (1881)
Animal Intelligence

Fig. 2.2. George Romanes

Historical Background

Romanes (1881)
Animal Intelligence

Insects ⟶ Birds ⟶ Apes ⟶ Humans

"They act with rational imitation – which involves a rational understanding of mechanical properties and considerable reasoning power"

Historical Background

Conway Lloyd Morgan
(1890s)

Fig. 2.5. Conwy Lloyd Morgan at about the time of his return to England

Historical Background

Conway Lloyd Morgan
(1890s)

Against
anthropomorphism

Fig. 2.5. Conwy Lloyd Morgan at about the time of his return to England

Historical Background

Conway Lloyd Morgan
(1890s)

Lloyd Morgan's (1894) Canon

In no case may we interpret an action as the outcome of the exercise of a higher psychical faculty, if it can be interpreted as the outcome of one which stands lower in the psychological scale

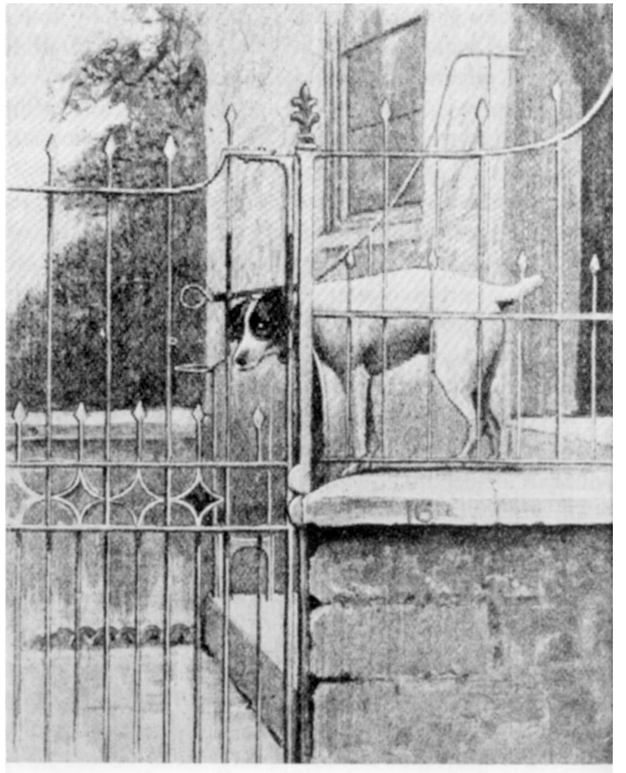


Fig. 2.6. A product of trial-and-error learning: Morgan's dog, Tony, operating the latch of a gate

Historical Background

Edward Thorndike (1874 - 1949)

Wanted to deliver
"the coup de grace to the despised
notion that animals reason"

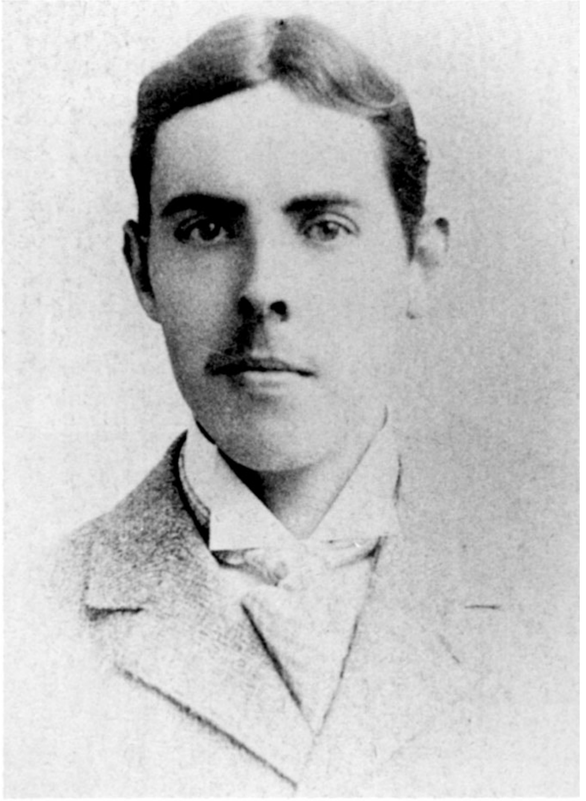


Fig. 3.7. Edward Thorndike on entering Harvard University.

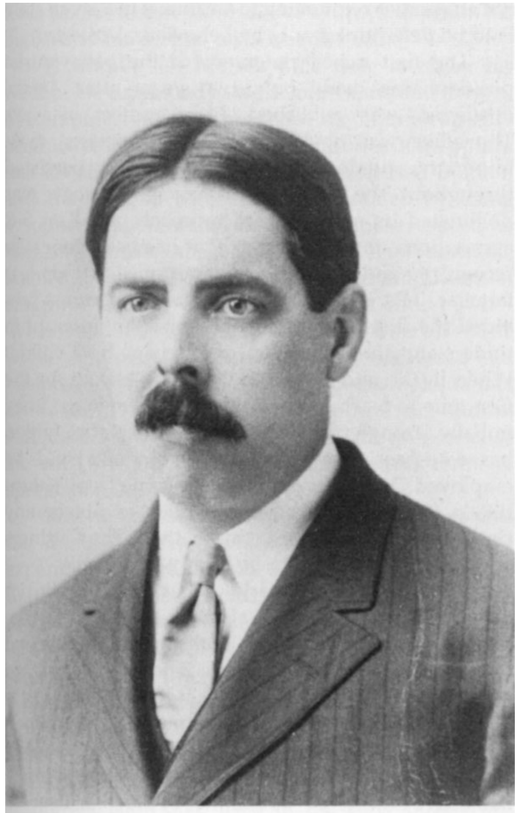


Fig. 3.10. Edward Thorndike in 1910, just prior to christening the Law of Effect

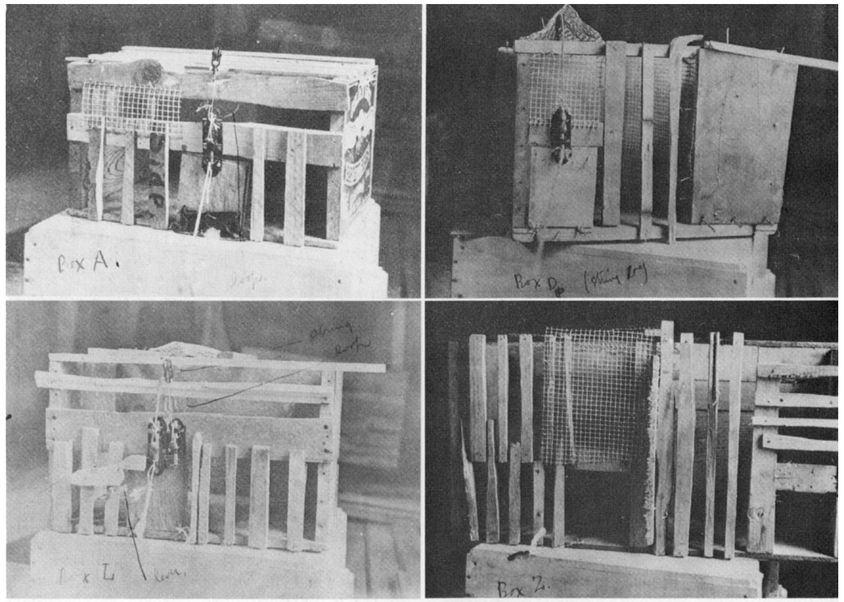
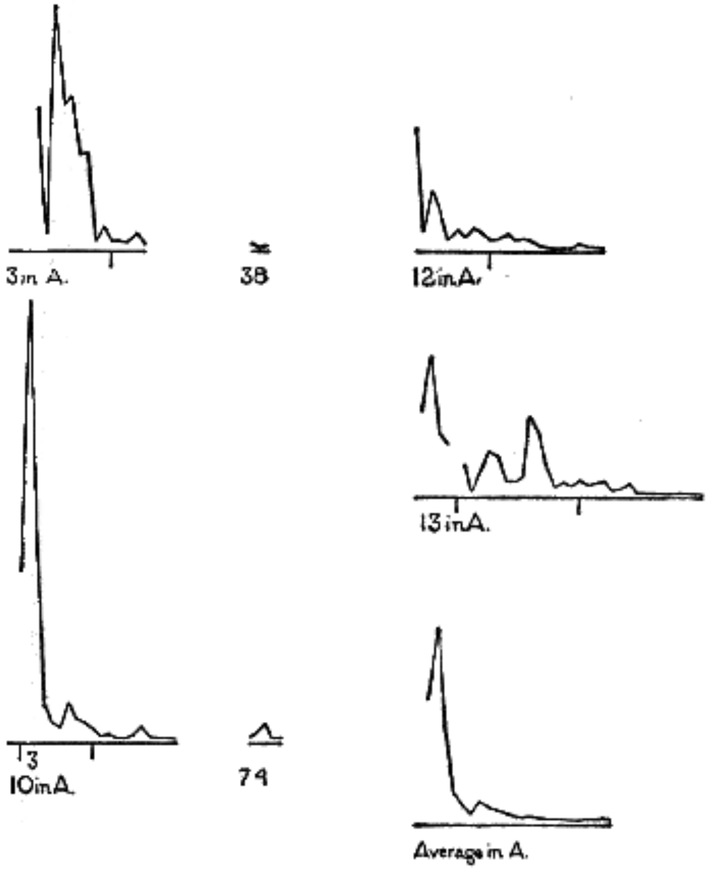
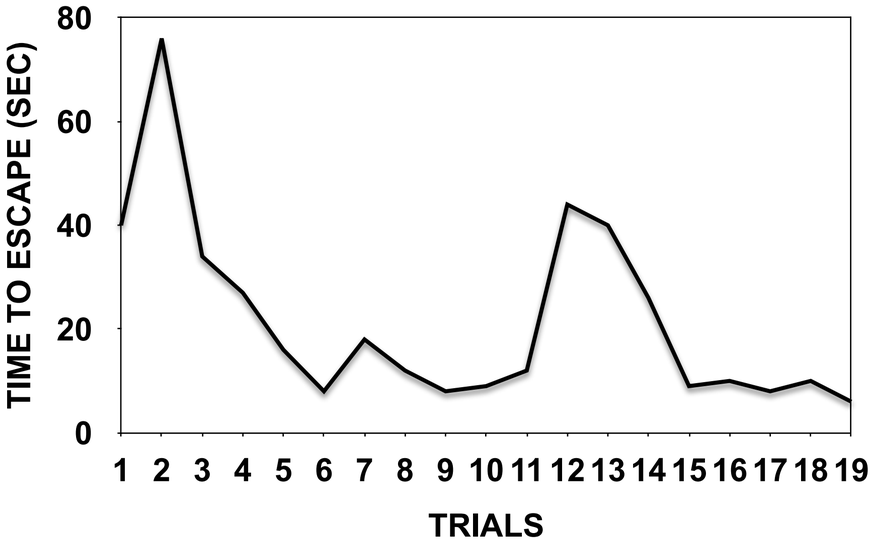


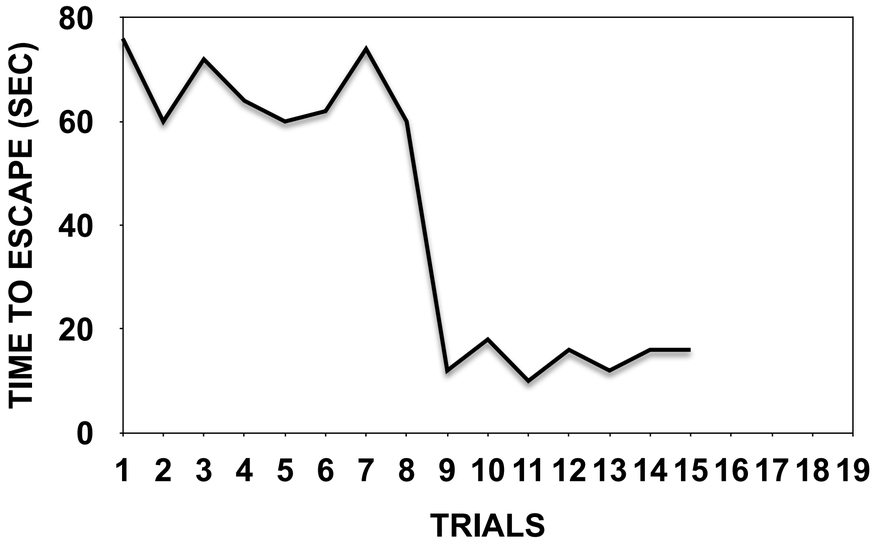
Fig. 3.8. Four of the puzzle boxes used by Thorndike in the research for his doctoral thesis



Trial by Trial Learning Curve

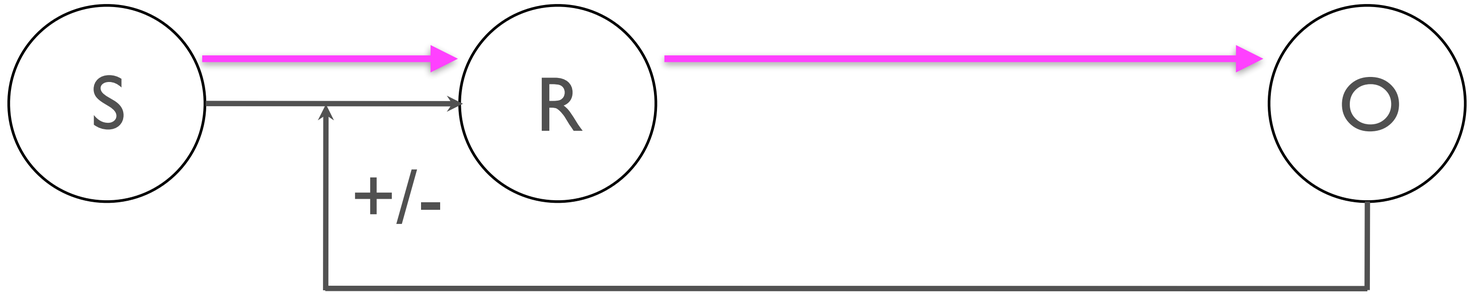


Hypothetical Learning Curve



Thorndike's law of effect

If a response leads to a satisfying outcome
it will be strengthened



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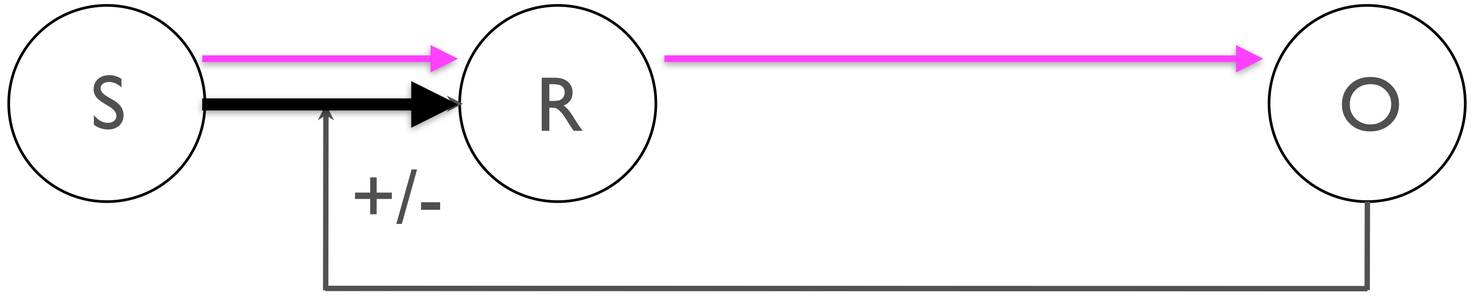

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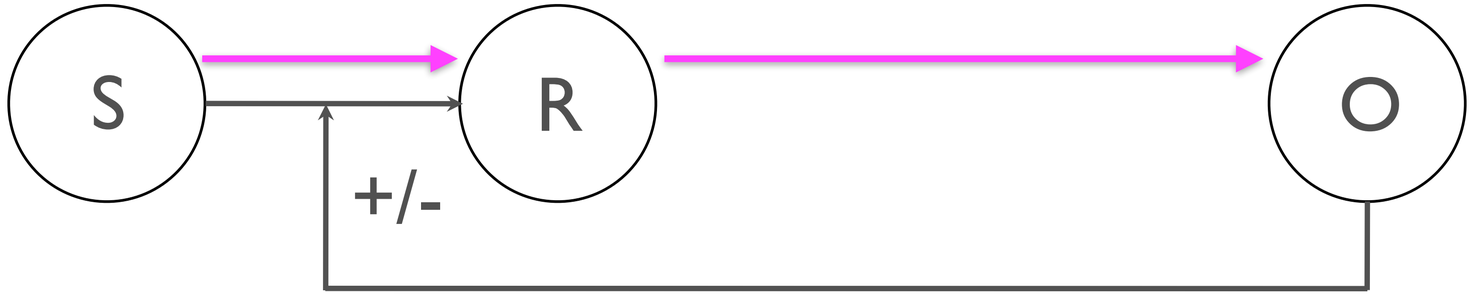

Historical Background

- ✦ John B Watson (1878-1958) advocated
  - Importance of learning
  - Behaviourism
  - Use of animals
- ✦ Burrhus Frederick Skinner (1904-1990)
- ✦ Ivan P Pavlov (1849-1936)

Watson

Titchener

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"Give me a dozen healthy infants …. and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select - doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors."

Watson

Rosalie Raynor

Little Albert Experiment (1920)

A Clockwork Orange (1971)

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